THIRD REVIEW REPORT

Project Title: AI-Based Smart Dustbin Monitoring System

1. Project Overview

The Smart Dustbin project is an AI-based waste management solution developed to improve waste monitoring and collection efficiency. The system aims to reduce overflowing dustbins, improve waste management, and minimize manual monitoring through an intelligent software application.

The project provides a web-based platform to monitor dustbin information and manage waste-related data efficiently.

2. Project Development and Implementation

1. Problem Identification and Analysis

- Identified problems related to overflowing dustbins and improper waste management.
- Conducted a field study in Goundanur, Coimbatore.
- Observed issues such as midday dustbin overflow and the absence of proper wet and dry waste segregation.
- Defined the project objectives and proposed an AI-based solution.

2. System Design

- Designed the complete system architecture.
- Planned the integration of AI-based waste detection, database management, and dustbin monitoring.
- Designed the user interface and dashboard.

3. Software Development

- Developed the Smart Dustbin web application using Python and Flask.
- Implemented the backend functionality.
- Created a database to store and manage dustbin information.
- Developed an AI-based waste detection module.
- Implemented an ESP32 simulator for software-based testing.
- Integrated frontend and backend components.

4. Website Development

- Successfully developed the complete Smart Dustbin website.
- Created an interactive dashboard for displaying dustbin information.
- Integrated the application modules.
- Designed the interface to make monitoring and accessing information easier.

5. Deployment and QR Code Generation

- Successfully deployed the website on Render.
- Verified the availability of the deployed application.
- Generated a QR code for accessing the website easily.
- The QR code provides direct access to the Smart Dustbin application.

3. Technologies Used

- Programming Language: Python
- Backend: Flask
- Frontend: HTML, CSS, JavaScript
- Database: SQLite
- AI Module: Python-based AI detection
- Hardware Simulation: ESP32 Simulator
- Deployment: Render
- Version Control: GitHub

4. Project Completion Status

The complete software project has been successfully developed and deployed. The frontend, backend, database, AI detection module, and dashboard have been implemented and integrated.

The website is accessible online, and a QR code has been generated for convenient access.

5. Project Outcome

- Successfully developed a functional Smart Dustbin web application.
- Implemented AI-based waste detection functionality.
- Developed a monitoring dashboard.
- Integrated database functionality.
- Successfully deployed the website online.
- Generated a QR code for easy website access.

6. Conclusion

The Smart Dustbin project has been successfully completed as a software-based solution. The project addresses real-world waste management challenges through AI-based detection and a web-based monitoring platform.

The successful development and deployment of the website demonstrate the practical implementation of the proposed solution. This project helped in gaining practical experience in Python programming, AI integration, database management, web development, and application deployment.